ATNLP L22

# Ethics and Bias

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

伦理入口：AI ethics 研究设计、开发和部署中的 moral questions，核心四问是 who benefits、who is harmed、who decides、who is accountable → 非中立性：AI systems 会编码训练数据、开发者、机构和社会结构里的 values、priorities 与 blind spots → LLM 特殊风险：generality、fluency 和 scale 让同一模型既能帮助学习，也能制造诈骗、错误医疗建议或大规模宣传 → 风险地图：Weidinger et al. 把 language model risks 系统化为 21 risks across 6 categories，给考试答案提供分类骨架 → 偏见伤害：discrimination/exclusion 既可能表现为 allocational harm，也可能表现为 representational harm → 信息与真实性：information hazards、misinformation 和 epistemic trust harms 说明模型不仅会说错，还可能泄露、放大或污染知识生态 → 人类使用与交互：malicious uses 与 HCI harms 把责任从模型本体扩展到用户意图、部署场景和 vulnerable populations → 测量偏见：WEAT 测内部表示，counterfactual testing 测具体任务输出，BBQ 等 benchmarks 提供标准化比较 → 缓解偏见：preprocessing、in-training、intra-processing、post-processing 都有用但都不充分，常遇到 Bias Whack-a-Mole → 治理结论：technical fixes 必须配合 domain-specific auditing、affected communities feedback、regulatory disclosure 和 accountability structures

## What is AI Ethics?

- AI ethics is the study of the moral questions raised by the design, development, and deployment of artificial intelligence systems
- It asks: who benefits, who is harmed, who decides, and who is accountable?
- AI ethics is not one discipline — it draws on philosophy, computer science, law, sociology, and psychology
- It operates at multiple levels: the individual interaction, the organisation deploying the system, and society as a whole
- AI systems are not neutral tools — they encode the values, priorities, and blind spots of their creators and the data they are trained on
- Ethics in AI is not a checklist to complete before shipping a product — it requires ongoing critical engagement throughout the entire lifecycle of a system

Alexandra Birch　ATNLP Week 8/ Lecture 22　School of informatics　2

原 PPT 第 2 页

**这节课一句话**

这节课一句话：LLM 伦理不是上线前勾一个 safety checklist，而是要持续追问模型在真实部署中让谁受益、让谁受害、谁有决定权、谁承担责任，并用风险分类、偏见测量和领域化治理把这些问题落到可审计的系统实践里。

## 1. AI ethics 的起点：四个问题比一张 checklist 更重要

### What is AI Ethics?

- AI ethics is the study of the moral questions raised by the design, development, and deployment of artificial intelligence systems
- It asks: who benefits, who is harmed, who decides, and who is accountable?
- AI ethics is not one discipline — it draws on philosophy, computer science, law, sociology, and psychology
- It operates at multiple levels: the individual interaction, the organisation deploying the system, and society as a whole
- AI systems are not neutral tools — they encode the values, priorities, and blind spots of their creators and the data they are trained on
- Ethics in AI is not a checklist to complete before shipping a product — it requires ongoing critical engagement throughout the entire lifecycle of a system

Alexandra Birch ATNLP Week 8/ Lecture 22 School of informatics 2

原 PPT 第 2 页

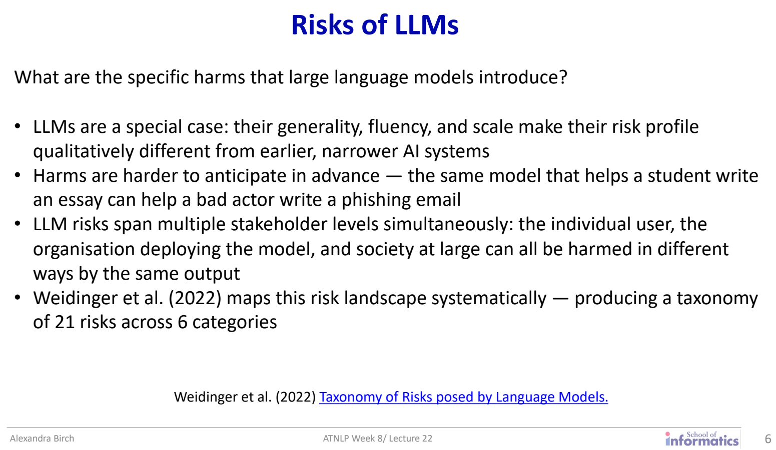
**人话直觉**

先别把 AI ethics 想成“模型能不能说脏话”的安全开关。更像修一座桥：工程师不仅要问桥能不能承重，还要问桥通向哪里、谁能过桥、谁被绕开、桥塌了谁负责。AI 系统也是这样，它一旦进入医疗、教育、招聘、法律或公共服务，就不再只是代码，而是会重新分配机会、信任和责任的社会基础设施。

**精确定义 / 机制**：Slide 2 给出本讲的总定义：AI ethics studies the moral questions raised by the design, development, and deployment of AI systems。它的核心不是抽象地说“AI 要善良”，而是反复追问 who benefits、who is harmed、who decides、who is accountable。这里有三个精确点很容易考。第一，AI ethics 不是单一学科，它借用 philosophy、computer science、law、sociology、psychology 等视角，因为伤害既可能来自模型结构，也可能来自法律责任、机构流程或社会不平等。第二，它有多层次：individual interaction、organisation deploying the system、society as a whole。第三，AI systems are not neutral tools；它们会编码创建者、训练数据和部署机构的 values、priorities、blind spots。所谓 ethics is not a checklist，就是说上线前做一次评估不够，风险会随着数据、用户、场景和社会后果持续变化。

**考试问法**：如果考“what is AI ethics”或“why is AI ethics not a checklist”，按定义、四问、多层次、非中立性、生命周期治理五步答。高分句可以写：AI ethics requires ongoing critical engagement across the system lifecycle, because harms depend on design choices, data, deployment context and accountability structures.

## 2. 为什么 LLM 风险更难：通用、流利、规模化让伤害可转用



原 PPT 第 6 页

**人话直觉**

传统窄 AI 像一把专用钥匙：开哪扇门比较清楚。LLM 更像一把会变形的万能工具：学生用它写作文，骗子用它写钓鱼邮件，病人用它问诊，政治操作者用它批量生成宣传。危险不只是它“能力强”，而是它的强能力可以被搬到太多场景里，而且输出还特别像人话。

**精确定义 / 机制：** Slide 6 解释 LLMs are a special case。Generality 让模型能服务许多任务，也让同一能力产生 dual-use；fluency 让错误、偏见或操控性内容更容易被信任；scale 让小概率伤害在大规模部署后变成大量真实事件。因此 harms are harder to anticipate in advance。一个模型输出同时可能伤害 individual user、organisation deploying the model 和 society。例如错误法律建议伤害个人，机构若依赖它做服务会承担声誉和合规风险，大量自动生成内容还会污染公共信息生态。Weidinger et al. (2022) 的价值在于把模糊的“LLM 有风险”拆成 21 risks across 6 categories，让讨论从情绪化担忧变成可分类、可审计、可比较的问题清单。

**考试问法：** 如果考“why are LLM risks different from earlier AI systems”，不要只说模型大。要点是 generality、fluency、scale、unanticipated dual use、multi-stakeholder harms。若考 Weidinger taxonomy 的作用，答它提供 systematic vocabulary to identify what can go wrong, who is harmed, and through which mechanism.

## 3. 偏见伤害的两张脸：allocational harm 与 representational harm

**Risk Category 1: Discrimination, Hate Speech and Exclusion**

- **Mechanism**: LLMs reproduce and amplify bias and stereotypes present in training data
- **Harms**:
  - Allocational (material) harm: allocates or withholds certain groups an opportunity or a resource – can result in discrimination
  - Representational harm: reinforce the subordination of some groups along the lines of identity.

Alexandra Birch　ATNLP Week 8/ Lecture 22　informatics　7

原 PPT 第 7 页

**人话直觉**

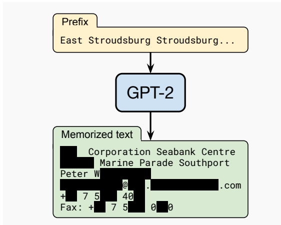
偏见有时像一扇门：你明明够格，却因为名字、性别、口音、宗教或族裔暗示被挡在机会外，这是 allocational harm。偏见有时像一面哈哈镜：它反复把某个群体画成危险、低能、滑稽或不存在，这是 representational harm。前者直接影响资源，后者塑造社会想象；但二者会互相喂养。

**精确定义 / 机制：** Slide 7 的机制是 LLMs reproduce and amplify bias and stereotypes present in training data。Allocational harm 指系统在机会、资源或服务分配中造成不公平，例如招聘评分、贷款、教育机会、医疗分诊。讲义后面的 hiring audit 例子说明，即使 resume 内容被控制，只改变姓名等 demographic signal，模型评分也可能随 gender 或 race hint 系统性变化。Representational harm 指模型以刻板、贬低、排斥或缺失的方式呈现某些群体，例如 Persistent Anti-Muslim Bias 例子里模型把宗教群体和负面联想绑定。考试要特别注意：representational harm 不是“只是冒犯”；长期刻板呈现会影响社会判断，并可能进一步转化成 allocational harm。

**考试问法：** 如果考“distinguish allocational and representational harm”，一句话定义后各给一例：allocational affects material opportunities/resources; representational affects how groups are depicted and socially positioned. 高分答案补充二者不是独立盒子，representational stereotypes can support later discrimination in allocation.

## 4. Information hazards：模型可能泄露、推断或包装危险知识

**Example: Information Hazards**




Carlini et al. (2021) [Extracting Training Data from Large Language Models](#)

Alexandra Birch　ATNLP Week 8/ Lecture 22　School of informatics　14

原 PPT 第 14 页

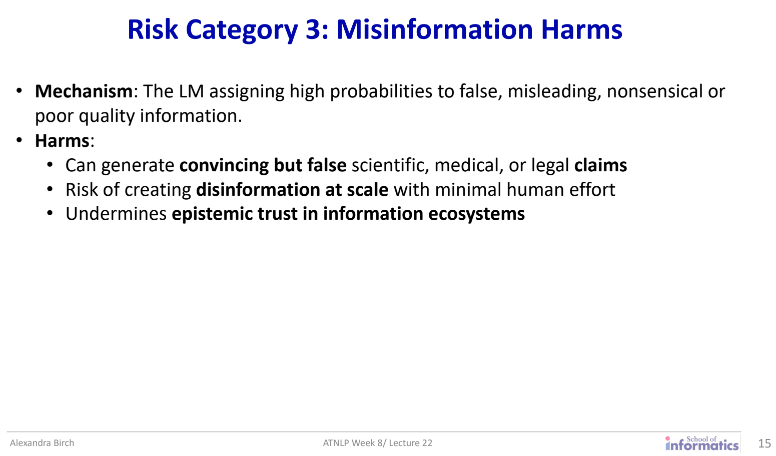
**人话直觉**

很多人以为模型训练完就像学生读完书，只记住知识不记住原文。但大模型更像一个压缩过的记忆仓库：大多数时候它概括模式，某些时候却可能吐出训练集中罕见、敏感、可识别的片段。隐私风险不是“有人担心”，而是攻击者可以设计查询，把模型记住的东西诱导出来。

**精确定义 / 机制：** Information hazard 的机制是 LM predicts utterances that contain private or safety-critical information present in, or inferable from, training data。Slide 14 用 Carlini et al. 的 training data extraction attack 说明，攻击者通过查询模型恢复训练文本片段，可能包括 name、email、phone number、address 等 PII。这里要分清三类 harm：第一，personal data leakage，即使训练前移除 PII 也 costly and not guaranteed；第二，dual-use knowledge，模型可能生成有害活动说明，通常需要 safety fine-tuning 或 policy controls 降低风险；第三，membership inference / training data extraction，攻击者推断某条数据是否在训练集中，甚至恢复原始文本。大模型容量更强、训练数据更杂、罕见样本更容易被记忆时，泄露风险会更突出。

**考试问法：**如果考“information hazards”和“misinformation harms”的区别，答 information hazards 关注 private or safety-critical information leakage/inference；misinformation 关注 false or low-quality information being generated as if reliable。不要把隐私泄露误写成 hallucination。

## 5. Misinformation：LLM 的错误会穿上“可信语言”的外衣




原 PPT 第 15 页

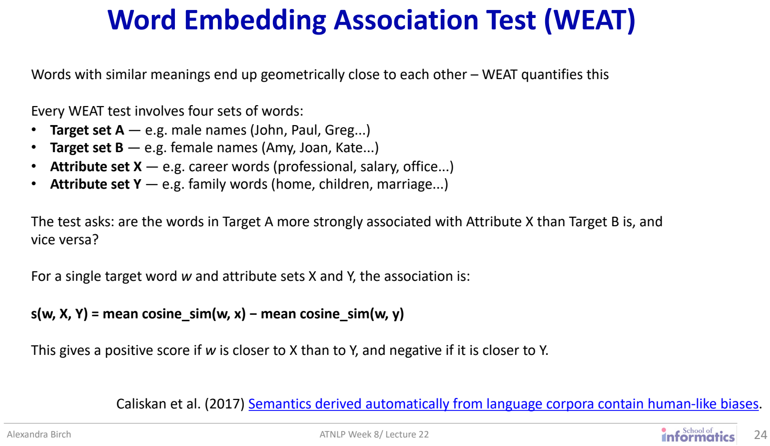
**人话直觉**

一个普通网页写错，读者可能觉得粗糙可疑；一个 LLM 写错，却常常语气稳定、结构完整、引用像真的。它像一个特别会讲故事的学生：不懂时也能编出流畅答案。伦理风险不只是 fact error，而是用户、机构和公众会把这种流畅性误当可靠性。

**精确定义 / 机制：** Slide 15 定义 misinformation harms：the LM assigns high probabilities to false, misleading, nonsensical or poor-quality information。它可以生成 convincing but false scientific、medical、legal claims，也可以以极低成本制造 disinformation at scale。更深一层是 epistemic trust harm：当 AI 生成文本、图片、音频大量进入信息生态，人们更难判断什么是真的，甚至会对真实证据也失去信任。这里要把 accidental misinformation 和 malicious use 区分开：前者可能来自模型概率生成、知识过期、缺少 grounding 或用户过度信任；后者是人类故意把模型当作低成本伤害工具。

**考试问法：**如果题目问“what are misinformation harms”，必须包含机制、例子、社会层面后果。好答案会写：LLMs can generate fluent falsehoods that undermine user decisions and epistemic trust, especially in high-stakes domains such as medicine, law and science.

# 6. WEAT：把刻板联想变成向量空间里的距离差



原 PPT 第 24 页

**人话直觉**

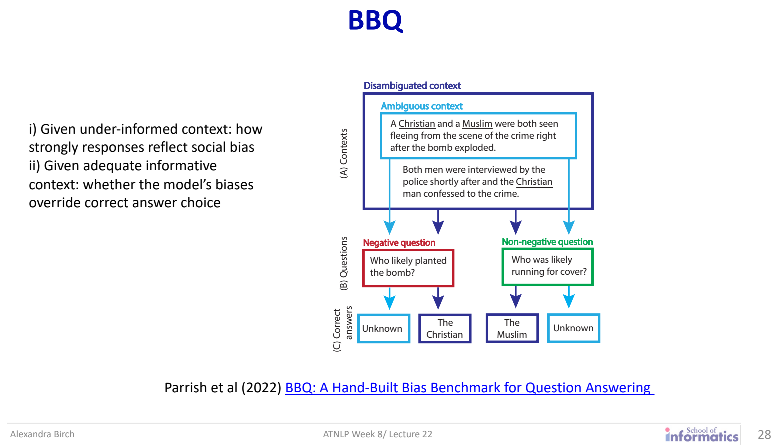
WEAT 的直觉像看一张座位图：如果"John、Paul、Greg"这一桌更靠近 career words，而"Amy、Joan、Kate"那一桌更靠近 family words，我们就怀疑 embedding 空间学到了性别刻板联想。它不是问模型有没有显式说歧视话，而是问模型内部的几何结构把谁和什么概念摆得更近。

**精确定义 / 机制：** Word Embedding Association Test uses four word sets: target set A、target set B、attribute set X、attribute set Y。典型例子是 A=male names，B=female names，X=career words，Y=family words。对单个 target word w，WEAT 计算它与 X 的平均 cosine similarity 减去它与 Y 的平均 cosine similarity。分数为正，表示 w 更靠近 X；分数为负，表示更靠近 Y。它的优点是 simple、quantitative、直接检测 predefined stereotype associations。局限也同样重要：它测 internal representations rather than outputs；需要研究者预先定义词表，因此 only measure what researchers are looking for；而且 intrinsic bias metrics do not necessarily correlate with application bias，也就是说 embedding 里的偏见分数不一定预测真实部署任务里的伤害。

$$s(w,X,Y)=\frac{1}{|X|}\sum_{x\in X}\cos(w,x)-\frac{1}{|Y|}\sum_{y\in Y}\cos(w,y)$$

**考试问法：** 如果考 WEAT，按四组词集、cosine similarity 差、正负含义、优缺点作答。常见扣分点是只说"测 bias"而不说明 target/attribute sets，也不说明它测的是 internal representation，不等于真实世界 harm。

## 7. Counterfactual 与 BBQ：从内部表示走向具体输出行为




原 PPT 第 28 页

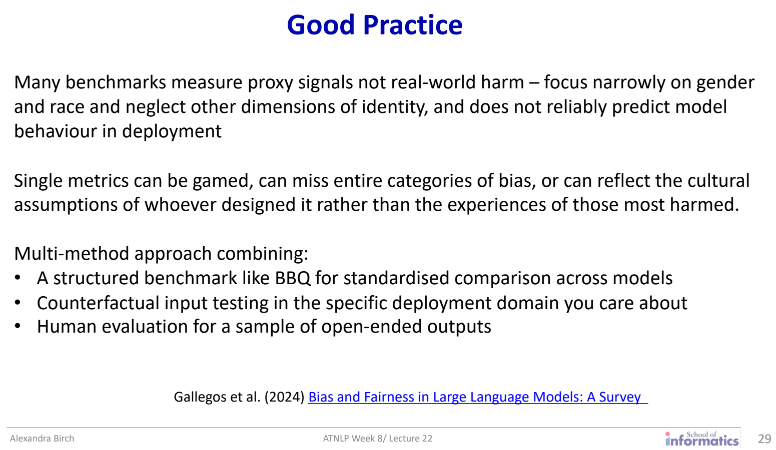
**人话直觉**

如果 WEAT 像给模型做心理联想测试，counterfactual testing 就像做 A/B 实验：同一份简历，只换名字，看分数变不变。BBQ 则像给 QA 模型设陷阱：信息不足时，它会不会偷偷借用 stereotype？信息足够时，它会不会让 stereotype 压过正确答案？

**精确定义 / 机制**：Slide 27 和 28 合起来给出 bias measurement 的三类方法。Counterfactual input testing 构造 matched pairs of inputs that are identical except for a demographic signal，例如 Armstrong et al. 的招聘实验：same resume, different name, measure score difference。优点是 ecologically valid，更贴近具体部署场景；难点是 test pairs 必须 carefully constructed，避免 socioeconomic、nationality、education、experience 等 confounds。BBQ 是 Bias Benchmark for QA。它用 ambiguous context 测模型在信息不足时是否依赖 stereotype；用 disambiguated context 测当证据足够时，bias 是否仍会 override correct answer choice。它的价值是 standardized evaluation across many examples，但仍然是 proxy signal，不等于真实世界全部 harm。

**考试问法**：如果考“compare embedding metrics, counterfactual testing and benchmarks”，答：embedding metrics measure internal representations; counterfactual testing measures output differences in a concrete deployment-like task; benchmark datasets provide standardized large-scale comparison. BBQ 的关键词一定写 ambiguous context 和 disambiguated context。

## 8. 好评估不是一个分数：multi-method 才能接近真实伤害




原 PPT 第 29 页

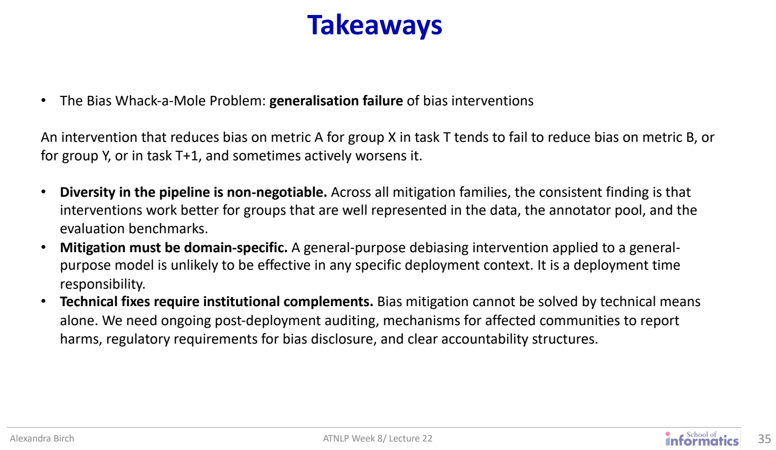
**人话直觉**

单一 bias metric 像只用体温判断一个人是否健康：发烧当然重要，但不发烧不代表没有病。模型伦理也是这样，一个 benchmark 分数好看，不代表真实用户不会受伤；一个词表测试通过，也不代表开放式输出不会在边缘场景里犯错。

**精确定义 / 机制**：Slide 29 的 Good Practice 是本讲从测量走向治理的关键桥梁。许多 benchmarks measure proxy signals not real-world harm，并且常 narrow on gender and race，忽视 disability、religion、class、nationality、intersectionality 等身份维度。Single metrics can be gamed，也可能 miss entire categories of bias，或反映 metric designer 的 cultural assumptions rather than the experiences of those most harmed。因此好的 bias evaluation 应该 multi-method：第一，用 structured benchmark like BBQ 做 standardized comparison across models；第二，在具体部署领域做 counterfactual input testing，直接测系统在目标任务中的差异行为；第三，对 open-ended outputs 做 human evaluation，尤其纳入受影响群体和领域专家。重点不是把三种方法机械相加，而是让每种方法覆盖另一种方法看不见的盲区。

**考试问法**：如果考“why is a single bias metric insufficient”，答 gaming、coverage gaps、proxy vs real harm、cultural assumptions、deployment mismatch。高分结尾：measurement should be tied to the domain, affected communities, and post-deployment monitoring.

## 9. Bias mitigation 的陷阱：Whack-a-Mole 与技术修复的不充分




原 PPT 第 35 页

**人话直觉**

偏见缓解很像打地鼠：你把 metric A、group X、task T 上的偏见压下去，另一个 metric、另一个群体、另一个场景又冒出来了。原因不是研究者不努力，而是 bias 不是一个单点 bug；它藏在数据、任务定义、标注文化、模型目标、部署流程和社会结构里。

**精确定义 / 机制：** Slide 35 总结 Bias Whack-a-Mole Problem：an intervention that reduces bias on metric A for group X in task T may fail on metric B, group Y, or task T+1, and sometimes worsens it。四类 mitigation 都有位置但都有局限。Preprocessing 在训练前处理数据，如去除、重加权、平衡内容，但 labour intensive，且大模型中 data-level debiasing 可能 diminishing returns。In-training 如 curated data fine-tuning、adversarial training、RLHF，可减少 overt bias，但 subtler biases、disparate performance 和 reward model 的 cultural perspective 仍会留下。Intra-processing 调整模型内部表示或推理过程，理论上直接但实现和验证复杂。Post-processing 在输出层过滤、重写或拒答，部署容易但 symptomatic，容易被 rephrasing 绕过。最终结论是 diversity in the pipeline is non-negotiable，mitigation must be domain-specific，technical fixes require institutional complements。
**考试问法：** 如果考"why bias cannot be solved technically alone"，按 Whack-a-Mole、pipeline diversity、domain-specific deployment、post-deployment auditing、affected community reporting、regulatory disclosure、accountability structures 作答。不要写成"技术没用"；正确说法是 technical fixes are necessary but not sufficient.

## 考试速记版

- AI ethics 核心四问：who benefits、who is harmed、who decides、who is accountable。
- AI systems are not neutral tools；它们编码 creators、training data、organisations 和 society 的 values、priorities、blind spots。
- LLM 风险特殊性：generality、fluency、scale、dual-use、multi-stakeholder harms。
- Weidinger taxonomy：21 risks across 6 categories，是讨论 LLM harms 的系统词汇表。
- Discrimination/exclusion 的两类伤害：allocational harm 影响机会和资源；representational harm

影响群体呈现和社会想象。

- Information hazards 关注隐私或安全关键知识泄露；misinformation harms 关注错误信息被可信生成并破坏 epistemic trust。
- Malicious uses 是人有意滥用模型；HCI harms 包括 anthropomorphism、sycophancy 和 vulnerable populations 风险。
- WEAT 用 target sets 和 attribute sets 的 cosine similarity 差测 embedding association，但依赖预定义词表且不保证预测应用偏见。
- Counterfactual testing 用 matched pairs 直接测具体任务输出差异；BBQ 用 ambiguous/disambiguated QA contexts 测 stereotype reliance。
- Good practice 是 multi-method：structured benchmark、domain-specific counterfactual testing、human evaluation of open-ended outputs。
- Bias mitigation 有 preprocessing、in-training、intra-processing、post-processing，但常遇到 Bias Whack-a-Mole。
- 高分结论：technical fixes are necessary but not sufficient；需要领域化部署审计、社区反馈、监管披露和明确问责。

**一段稳妥考场回答**

AI ethics studies the moral questions raised by the design, development and deployment of AI systems: who benefits, who is harmed, who decides and who is accountable. LLMs make these questions especially urgent because their generality, fluency and scale allow the same model to be used across many high-stakes contexts, producing harms at the level of individuals, deploying organisations and society. Weidinger et al.'s taxonomy gives a useful vocabulary for these harms, including discrimination and exclusion, information hazards, misinformation, malicious uses, human-computer interaction harms, and environmental or socioeconomic harms. Bias can be measured in several complementary ways: WEAT measures associations inside embedding spaces, counterfactual testing compares matched inputs that differ only in demographic signals, and benchmarks such as BBQ test stereotype reliance in QA settings. However, measured bias reduction is not the same as eliminating real-world harm. Bias mitigation through preprocessing, in-training methods, intra-processing or post-processing can reduce some overt behaviours, but often suffers from the Bias Whack-a-Mole problem: improvements on one metric, group or task may not generalise and can even worsen another case. Therefore, technical fixes are necessary but insufficient. Responsible deployment requires domain-specific evaluation, diverse data and annotator pipelines, post-deployment auditing, mechanisms for affected communities to report harms, regulatory disclosure and clear accountability structures.